

CSE 201 1. Midterm

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I. QUESTION (20 POINTS)

Design a class named `LibraryCard` to represent a library card owned by a student. The class contains:

- A string data field named **owner** that specify the owner of the card.
- An integer data field named **borrowCount** that specify the number of books the student has borrowed.
- A no-arg constructor that sets the `borrowCount` to 0.
- One argument constructor **`LibraryCard(String newOwner)`**.
- A method named **`borrowBooks(int numOfBooks)`** where exactly `numOfBooks` books are borrowed.
- A method named **`returnBooks(int numOfBooks)`** where exactly `numOfBooks` books are returned. Write also safety check code.
- Two getter methods for the two fields **owner** and **borrowCount**.

II. QUESTION (12 POINTS)

Write a main function using Question 1 to do the following:

- Declare three `LibraryCards` one owned by no one, the other owned by Seda, and the last one owned by Olcay.
- The following events occurred. Olcay borrowed 5 books, Seda borrowed 2 books, Olcay returned 2 books.
- Print the number of books borrowed by Olcay.
- Print the number of books borrowed by Seda.

III. QUESTION (12 POINTS)

Explain the change that need to be done in the definition of class `LibraryCard` so that we can easily find out the total number of books borrowed from the library.

IV. QUESTION (12 POINTS)

Write a function

`int notBorrowed(LibraryCard[] cards)`

which returns the number of students who haven't borrowed any books. The library cards of the students are stored in the `cards` array.

V. QUESTION (10 POINTS)

Write a function

`void exchangeBooks(LibraryCard student1, LibraryCard student2)`

which swaps the books borrowed by two students. The students exchange the books borrowed by them.

VI. QUESTION (20 POINTS)

Design a class named `Student` that contains:

- A string data field **name** for the student's name.
- An integer data field **id** for the student's id number.
- A string field **gender** for the student's gender.
- An integer variable **numOfStudents** to count the students.
- A constructor that creates a student with specified name, id, and gender.
- The accessor methods for name and gender.
- The mutator methods for id.

VII. QUESTION (14 POINTS)

Draw UML for the `Student` class defined in Question 6.